Basics of Pharmacogenomics

•Pharmacogenetics

•Study of the relationship between variations in a **single gene** and variability in drug disposition, response and toxicity

•Pharmacogenomics

•Study of the relationship between variations in a **multiple genes** (up to the whole genome)

What is Pharmacogenomics?

•It is the study of inherited **variations in genes** that determine a patient's **response to** a drug

•'Personalized medicine'

•Genotype

-Set of unique genes that determine specific trait in a person

•Phenotype

-An observable trait (outward expression) of genotype

Nucleotide → Gene → Chromosomes → DNA

•Nucleotide

•Basic structural unit of DNA and RNA

•Gene

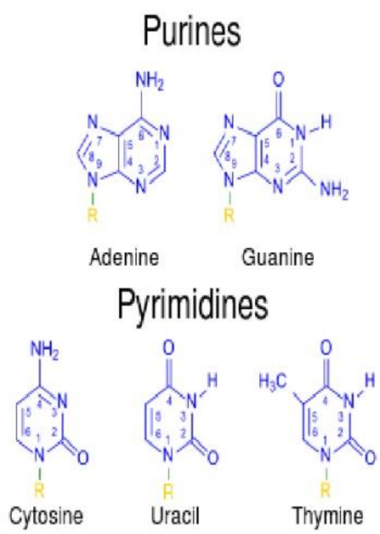
•A stretch of nucleotides that codes for a single protein

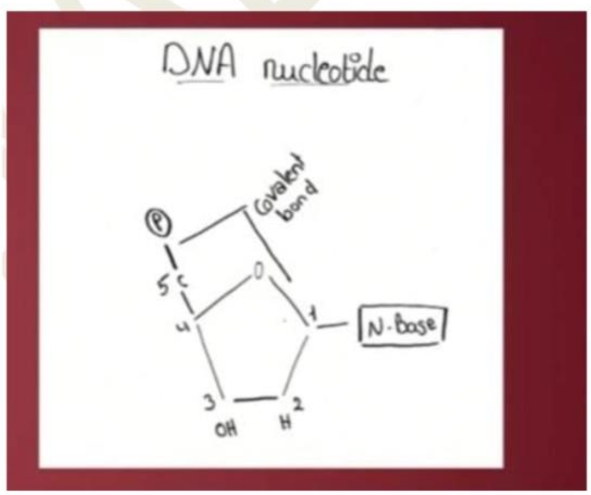
•Chromosomes

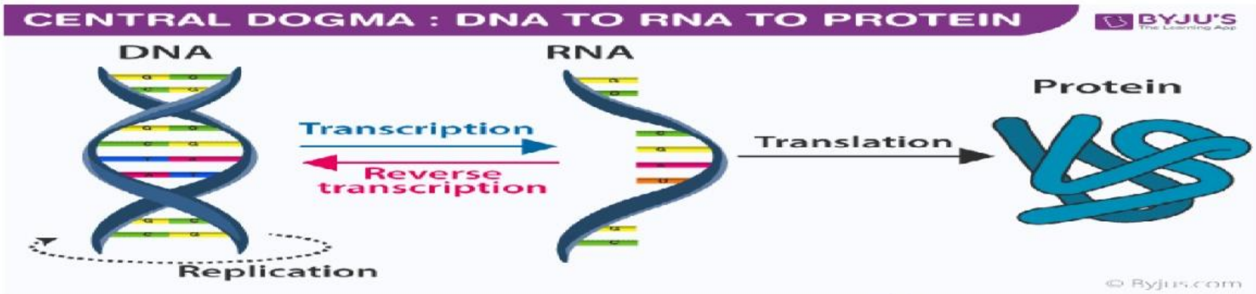
•Made of many genes that carry the genetic information

•DNA

•Genetic material that is the main component of (23 pairs)chromosomes (double helix)

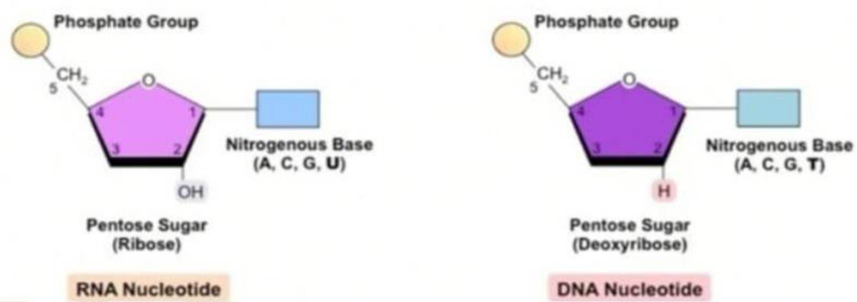


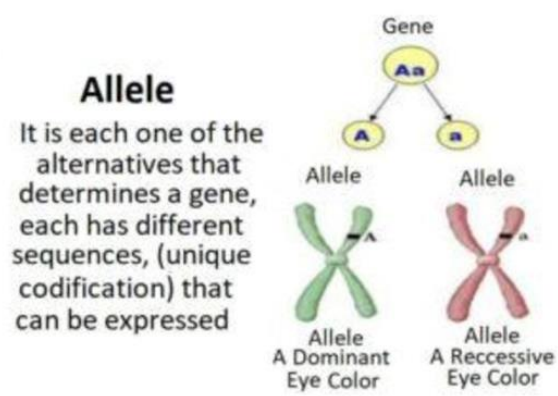




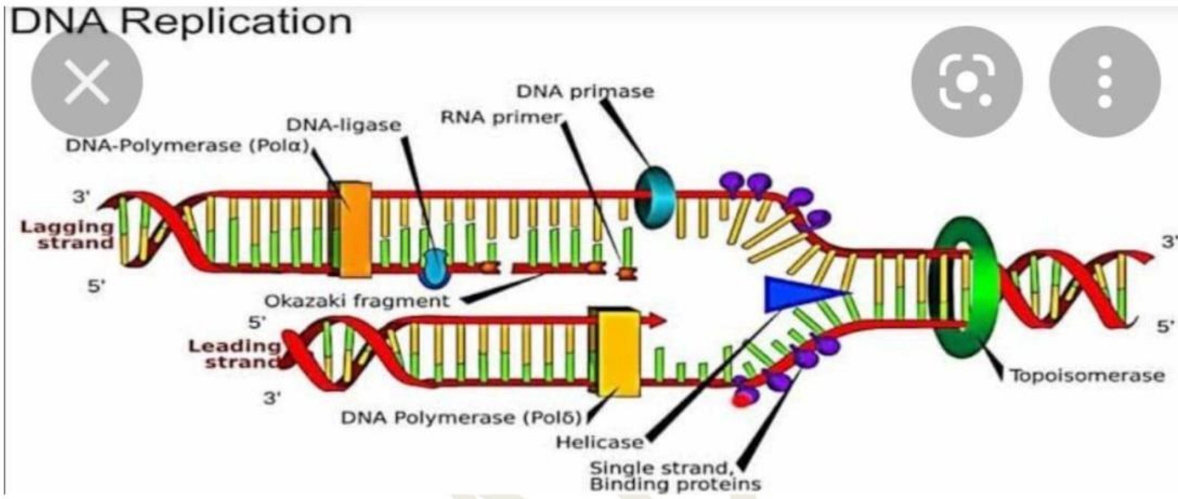
DNA and RNA

- The nucleotide units of both DNA and RNA are composed of:
  - A purine or pyrimidine base molecule
  - A sugar molecule (D-ribose in RNA or D-2-deoxyribose in DNA)
  - A phosphoric acid molecule





if alleles **identical** it call ? **Homozygous**
If **different** alleles ? **Heterozygous**



AUG *start code*
UUA UGA UAA *stop code*

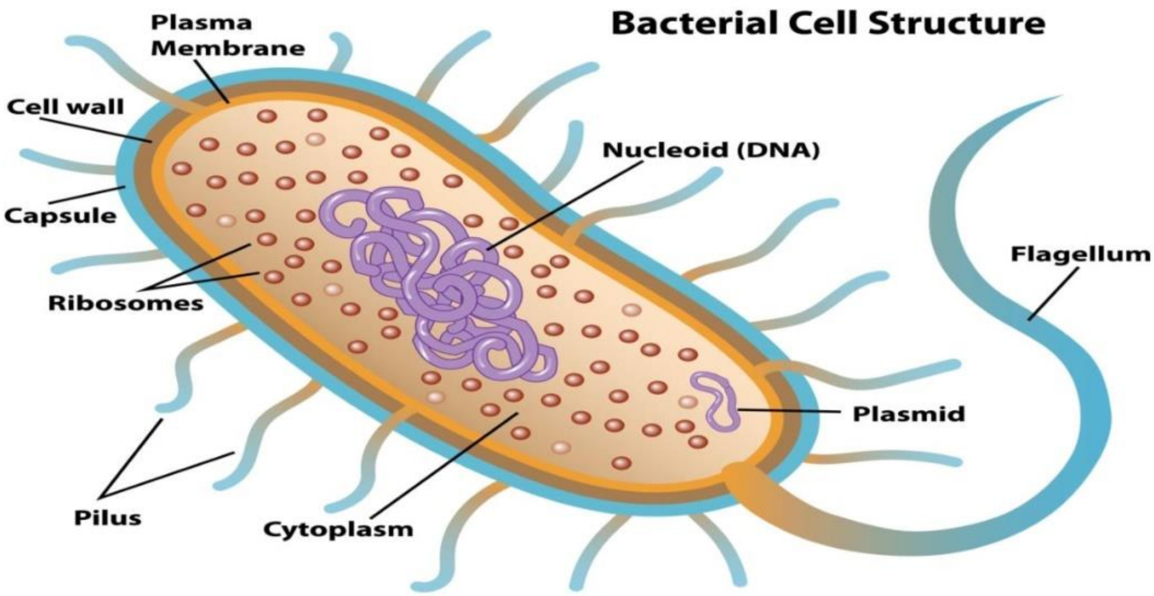
DNA supercoiling by??(topoisomerase enzyme)

Virus types are? DNA or RNA

Purine base?? A & G

لا يسمح بتداول أي جهد خارج
المشتركين في الكورس وتعتبر أمانه

P53 ? tumor suppressor

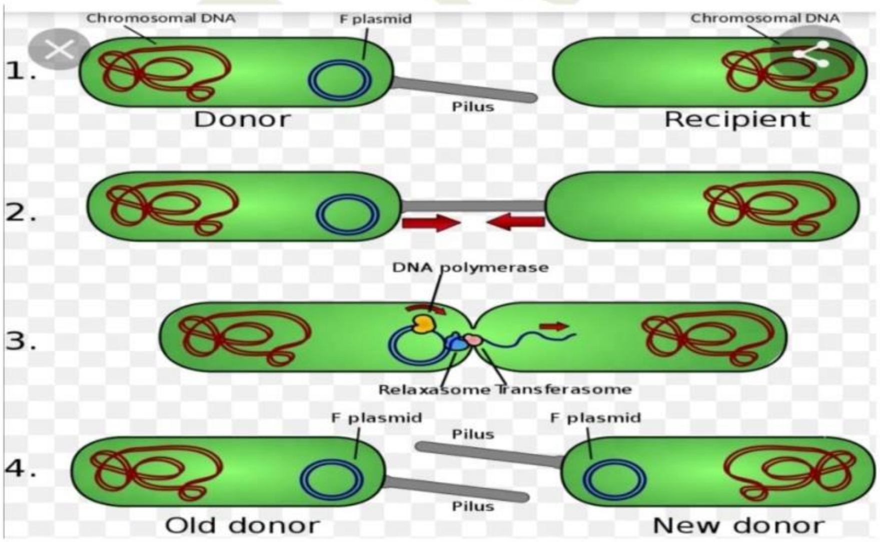


Bacteria

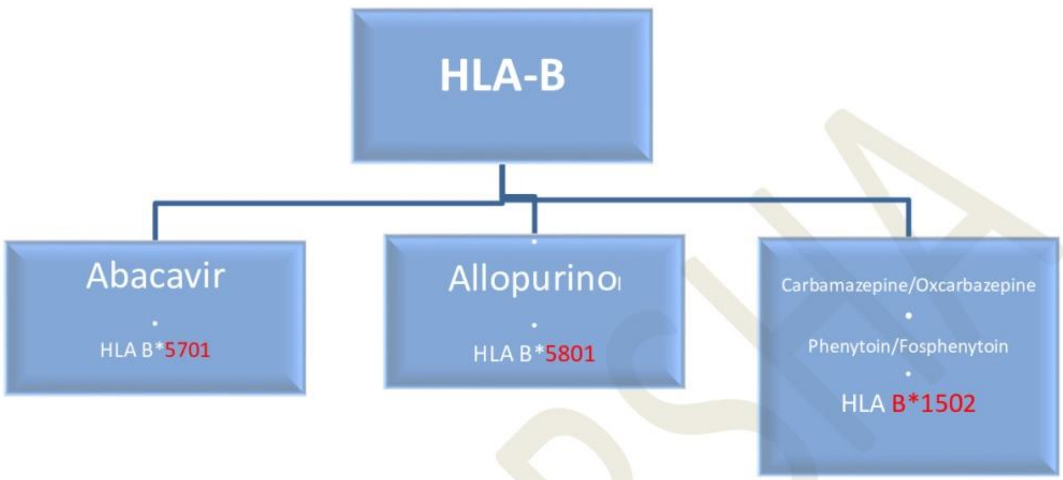
Pilli ? attachment & gene transfer during bacterial conjugation

Flagellum? Mainly for movement

Plasmid ?small circular double strand DNA ,, *antibiotic resistance*



1.Human leukocyte antigen (HLA), class I, B (HLA-B)



+ve → sever toxicity(hypersensitivity) st.jhonson syndrome(south asia at risk)

Avoid drug

2. CYP450 polymorphism

celecxoib → cyp2c9

Warfarin

Note cyp2c9*2 //cyp2c9*3, decrease warfarin metabolism

cyp2c9*1 >> normal

Vkorc1>>>> warfarin sensitivity & bleading

celecxoib cyp2c9*3,….. Decrease celecoxib metabolism

clopidogril (prodrug) → cyp2c19

voriconazole

clopidogril (prodrug) need cyp2c19*1 (normal) for activation

cyp2c19*2 //cyp2c19*3 bad (if present …no activation of clopidogrel)

- warfarin Vkorc1 **cyp2c9**
- **clopidogril** **cyp2c19**

Theophylline → **CYP1A2**

enzyme inhibitor

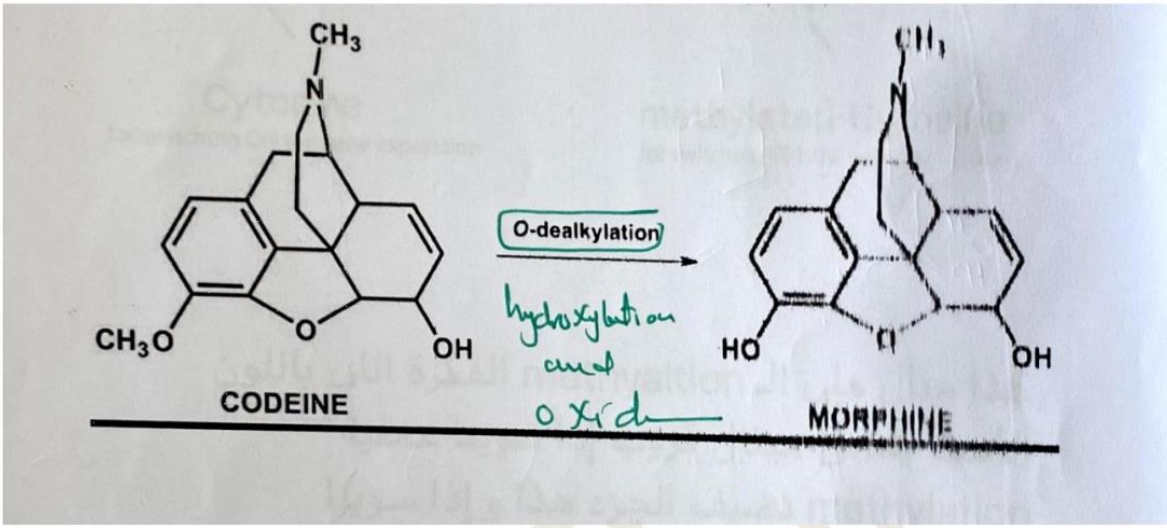
cyp --	cyp +
AB,antifungal ,omeprazole	rifampicin
Grape fruit, alcohol??	GPS…..grisofulvin..p….sulphonyl urea…
	st.jhon wart
Valporoic acid	carbamazepine ..phenytion
	Smooking

Alcohol-drug interactions differ not only with amount of alcohol taken but also with pattern of alcohol intake. When intake is **acute**, there is usually inhibition of the enzymes of drug metabolism. Alternatively, **chronic alcohol** abuse may lead to **enzyme induction**, increasing drug metabolism

Acute intake ⟶ ezyme inhibitor

Chronic intake ⟶ enzyme inducer

Codeine(antitussive) $\xrightarrow{\text{CYP2D6}}$ morphine (analgesic)



Rapid metabolizer??
Poor metabolizer??

Codeine

Tamoxiten $\xrightarrow{\text{CYP2D6}}$

Atomoxitine

Fluoxetine

Tramadol

TCA

• **Cetuximab&panitumumab** ⟶ **EGFR** ⊖

KRAS is G-protein in the EGFR pathway

KRAS mutation

+ve ⟶ no response ⟶ avoid use

HER2 gene

Trastuzumab & emtansine
pertuzumab & lapatinib
must be +ve to use this drugs

- Azathioprine
- 6-Mercaptopurine

- ↓ TPMT → ↑ Risk of **M**yelosuppression so must decrease dose

Thiopurine methyltransferase

Capecitabine
5-flurouracil

- ↓ **DPD** → Severe toxicity dihydropyrimidine dehydrogenase

- **D**iarrhea, **N**eutropenia and **N**eurotoxicity

- Irinotecan → UGT1A1*28 → • ↑ UGT1A1*28→ Neutropenia

UGT1A1 → Atazanavir
SLCO1B1 → Simvastatin
IL28B → PEG-Interferon

G6PD → ↓ G6PD → Hemolytic anemia

Summary :

• Avoid if positive (+) → HLA-B & KRAS
• Use if positive (+) → HER2

Questions

Pharmacogenomic testing shows a female patient with breast cancer is HER2 positive. She is likely to benefit from which of the following?

a. Ivacaftor

b. Leuprorelin

c. Trastuzumab

d. Rituximab

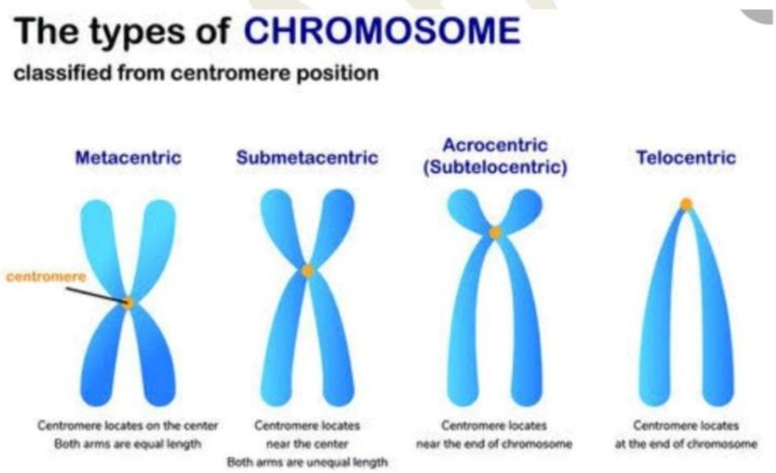
3- A child is rapid metabolizer of CYP2D6. How would his body respond?

a. Codeine would be rapidly metabolized to morphine

b. The child will have reduced opioid response

c. Codeine will not be converted into morphine

d. The child will have severe diarrhea



Chromosomes 13, 14, 15, 21, and 22 are?

A. Acrocentric

B. Heterozygous allele

C. Aneuploid

4- A Chinese patient is going to be started on carbamazepine. He should be tested for:

a. HLA-B*1501 allele

b. HLA-B*1502 allele

c. He does not need pharmacogenomic testing as he is from Asian descent

d. A gene because if he is found to have at-risk allele, he is more likely to experience agranulocytosis with the use of carbamazepine.

5- A patient is found to have been lacking in functional CYP2C19 enzyme activity. What would be your recommendation?

a. He should not receive Clopidogrel

b. He should not receive any CYPC19 enzyme inhibitors such as omeprazole or fluoxetine

c. If clopidogrel is administered, he will have increases INR

d. He is an ideal candidate to receive Clopidogrel

6- Which of the following best describes chromosome?

a. Building blocks of DNA, composed of four bases: adenine (A), Guanine (G), Thiamine (T) and Cytosine (C)

b. Organized into 23 pairs (46 chromosomes) as a supercoil structure

c. A stretch of DNA that codes for a single protein

d. Double helix molecule containing noncovalently bonded nucleotides

Polymorphism

Genetic variation is **common** 1% or more,,Genetic variation is inherite

Mutation
Genetic variation is rare less than 1%

Which antiviral c combination can be applied to all genotypes?

Sofospuvir and velpatasvir

Warfarin drug interaction?

CYP2C9

Clopidogrel drug interaction?

CYP2C19

Abacavir whats gene Of it?

HLB b 5701

What is two component of purine nitrogen base ?

Adenine , Guanine

What is the basic component of the cell?

Cell membrane or plasma, cytoplasm and nucleus

drug interactions between clopidogrel and omeprazole through which enzyme ?

CYP2C19

Pharmacogenomics of cetuximab?

KRAS mutation (-ve)

allopurinol genetic is HALA_B will be?

make. side effect of sever cutaneous

Carpemazinegenitic HALA_B?

epidemic necrosis

Which of the following is recombint DNA product?

Liragluaide

Clopidogrel interaction omeprazole Omeprazole and other PPI are Strong CYP2C19 inhibitors?

reduce antiplatelet effect

Abacavir gene HLA-B 5701

Clozapine + smoking Decrease clozapine plasma

what is cytochrome subtype responsible for drug interaction between omeprazol and clopidegrol ? CYP2C19

genetic test for azathioprine ?

TPMT

Abacavir test and which class?

HLAB 5701

Question about role of RNA (snRNA)

Types of RNA

RNA Type	Function
mRNA (mature RNA)	Shuttles the genetic code from DNA to ribosomes for translation
tRNA (transfer RNA)	Helps synthesize protein by providing a source of amino acids
rRNA (ribosomal RNA)	Associates with proteins to form ribosomes where proteins are translated
snRNA (small nuclear RNA)	Combine with proteins to form snRNPs to process premature RNA into mRNA

What the function of small nuclear (snRNAs) in protein synthesis ?

A)Act as catalyst

B)Modifies mRNA molecules

C)Genetic blueprint for the protein

D)Translates genetic code to amino acid

CYP2D6 with codeine when the effect is reduce?

Poor metabolizer

Which is purine in DNA?

purine [A,G]

DNA?

deoxyribonucleic acid

Trastuzumab action on breast cancer on which?

HERE2

Codeine CYTP2D6 poor metabolizer?

suffer more pain

Which one associated with genetic polymorphism?

Warfarin

Which one is tumor suppression protein?

P53

Case abute Epilepsy patient and postive HLA-B (1502) Which the following antiepileptic can be use ? A- Levtricetam ✅

B- Lamotrgin

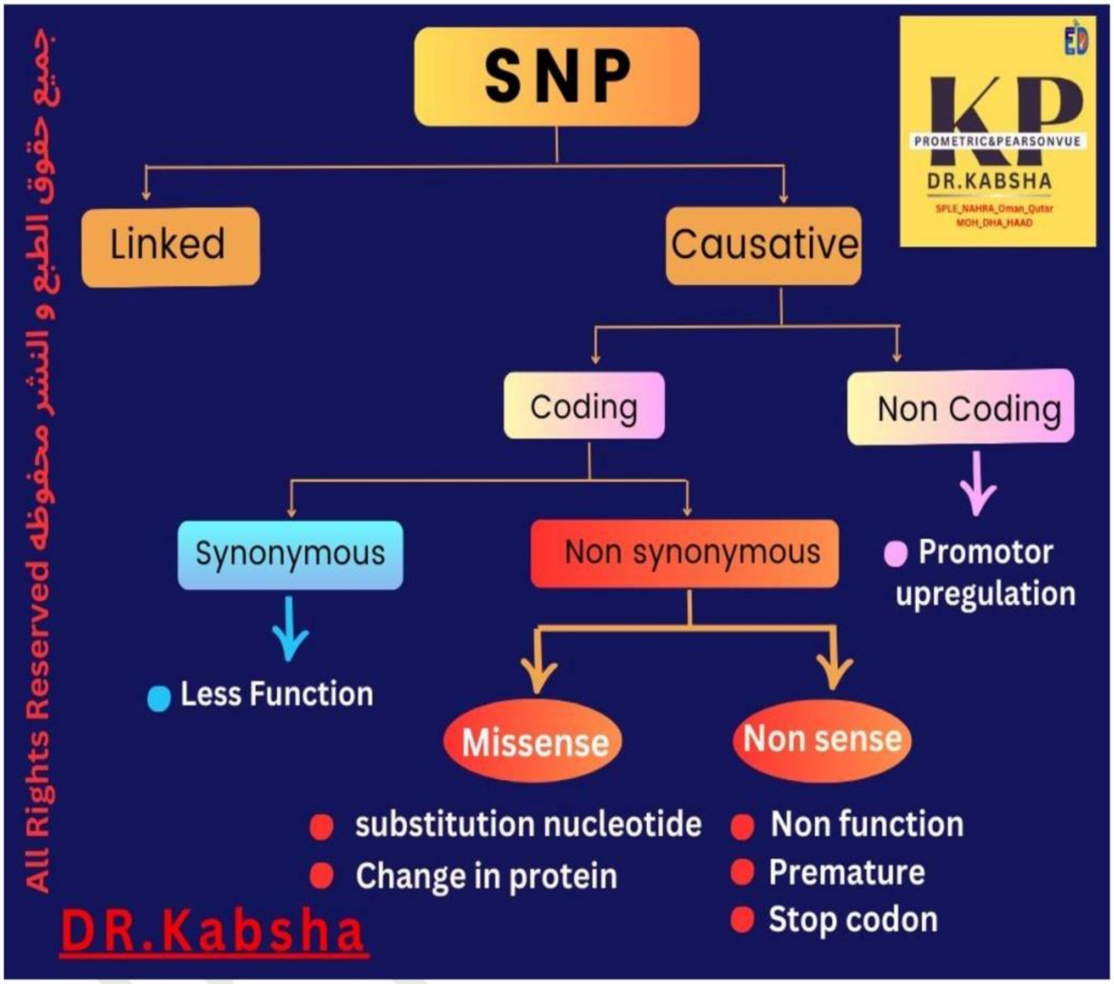
C- Phenytoin

D- Carbamazepine

Breast cancer.the gene HER2 was negative we can use ?

A- Letrozole ✅

B- Trustuzumb


SNP
Linked
Causative
Coding
Non Coding
Synonymous
Non synonymous
Less Function
Promotor upregulation
Missense
Non sense
substitution nucleotide
Change in protein
Non function
Premature
Stop codon
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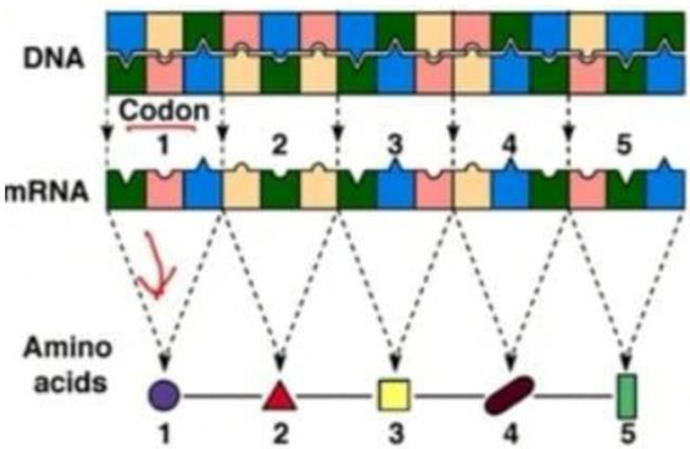
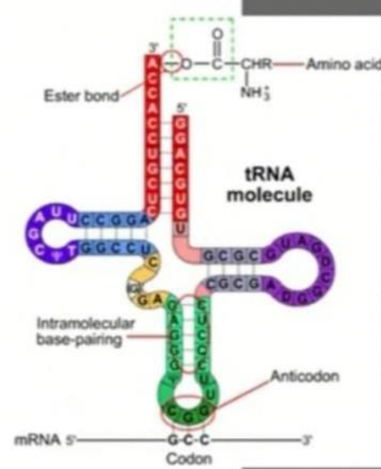
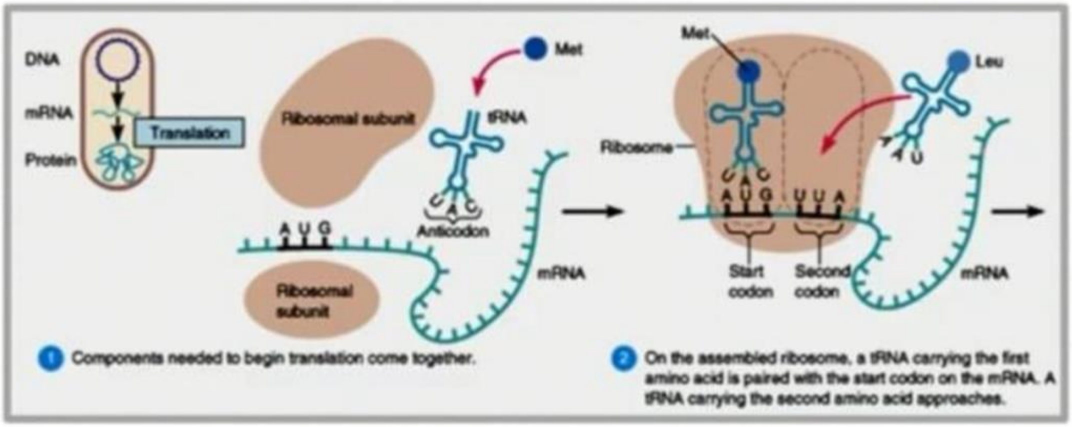


RNA and its different forms

- *Transfer RNA* (tRNA) is the highly ordered sequence of nucleotides folded into a cloverleaf like, inverted L structure with internal base-paired secondary structure.
- Typically, the three-nucleotide **anticodon** (complementary to the mRNA codon) is at the bottom of the inverted L.
- The 3'0H end of tRNA with a protruding ACCA is the location for adding the specific amino acid (in an aminoacyl bond) for that tRNA




DNA
mRNA
Protein
Translation
Ribosomal subunit
Met
tRNA
Anticodon
Ribosomal subunit
mRNA
Met
Leu
Ribosome
Start codon
Second codon
mRNA
1 Components needed to begin translation come together.
2 On the assembled ribosome, a tRNA carrying the first amino acid is paired with the start codon on the mRNA. A tRNA carrying the second amino acid approaches.